

Determining Solutions to InequalitiesAdditional Practice 7A.01.08 **ANSWER KEY**

Use this table of possible values for questions 1 and 2.

-11.5	36	$6\frac{3}{8}$	-8	-0.1212
-25	-9.5	-7.75	$6\frac{2}{7}$	-17.3

1. Which values from the table are solutions to $-3 + x/2 < -7$?

$-3 + x/2 < -7$ means $x/2 < -4$, so $x < -8$.

Solutions: -25, -17.3, -11.5, -9.5

2. Which values from the table are solutions to $-3 + x/2 > -7$?

$-3 + x/2 > -7$ means $x > -8$.

Solutions: -7.75, -0.1212, $6\frac{2}{7}$, $6\frac{3}{8}$, 36

-8 is in neither list, because at $x = -8$ both sides are exactly equal.

3. Rewrite both inequalities so that -8 is a solution to each. Explain your reasoning.

$-3 + x/2 \leq -7$ and $-3 + x/2 \geq -7$

At $x = -8$ the left side equals -7 exactly, so -8 is the boundary. Adding the equal sign lets the boundary count.

4. Complete the sentence. An inequality has _____ solutions.

infinitely many

5. Marco and Elena were solving $2 - 0.5x > 6$. Marco says the solution is $x > -8$. Elena says the solution is $x < -8$. Do you agree with one of them, or with neither? Justify your answer mathematically.

Elena is correct.

$2 - 0.5x > 6 \rightarrow -0.5x > 4 \rightarrow x < -8$ (divide by -0.5 and reverse the symbol)

Check: $x = -10$ gives $2 + 5 = 7 > 6$ ✓, $x = 0$ gives $2 > 6$ ✗

Marco divided by -0.5 without reversing. Unlike the worksheet's question 6, one of these two students is right — read before you assume.

6. Complete the sentence. When you multiply or divide both sides of an inequality by a negative number, the inequality symbol _____.

reverses (flips to face the other way)

For questions 7 through 10, solve the inequality, then graph the solution set on the number line provided.

7. $-8 \leq a/(-3) - 9$

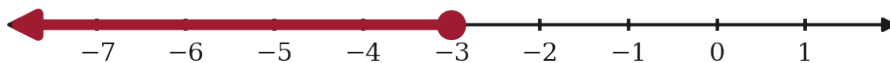
$1 \leq a/(-3)$ (add 9 to both sides)

$-3 \geq a$ (multiply both sides by -3 and reverse)

$a \leq -3$

Check: $a = -6$ gives $2 - 9 = -7$, and $-8 \leq -7$ ✓; $a = 0$ gives -9 , and $-8 \leq -9$ ✗

Solution. $a \leq -3$



$a/(-3)$ is the same as $-a/3$. Dividing by a negative denominator reverses the symbol exactly as dividing by a negative coefficient does.

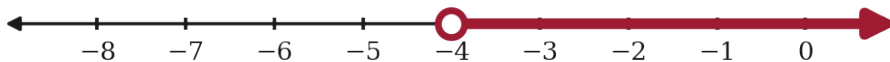
8. $9m + 16 > -20$

$9m > -36$ (subtract 16 from both sides)

$m > -4$ (divide both sides by 9)

Check: $m = 0$ gives $16 > -20$ ✓, $m = -4$ gives $-20 > -20$ ✗

Solution. $m > -4$



Dividing by a positive 9 leaves the symbol alone.

9. $2.8 < -1.3p - 0.5$

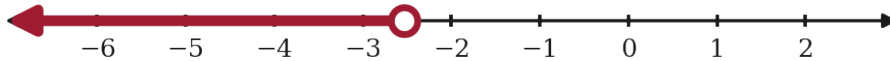
$3.3 < -1.3p$ (add 0.5 to both sides)

$-2.538... > p$ (divide both sides by -1.3 and reverse)

$p < -33/13 \approx -2.54$

Check: $p = -3$ gives $3.9 - 0.5 = 3.4 > 2.8$ ✓, $p = -2$ gives $2.6 - 0.5 = 2.1 > 2.8$ ✗

Solution. $p < -33/13$, about -2.54



The boundary is not a whole number. Place the open circle between -3 and -2 , a little past halfway toward -3 , and say in words which two integers it falls between. On the FAST an estimate placed between the correct pair of integers is what is being asked for.

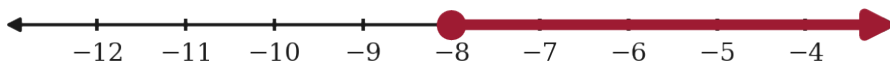
10. $-2 + (5/4)x \geq -12$

$(5/4)x \geq -10$ (add 2 to both sides)

$x \geq -8$ (multiply both sides by $4/5$)

Check: $x = -8$ gives $-2 - 10 = -12 \geq -12$ ✓, $x = -12$ gives $-2 - 15 = -17 \geq -12$ ✗

Solution. $x \geq -8$



Multiplying by the reciprocal $4/5$ is positive, so the symbol stays put.